

CineEmbed

Multi-Modal Unsupervised Embedding of 329,044 Films

Intermediate Progress Report • v1.0

Ahmet Baran Dincoguz • Arda Arvas • Bertan Kaan Kaya

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Repository: github.com/barandincoguz/CineEmbed

Executive Summary

COMPLETE Modeling MVP phase delivered on schedule. The intermediate phase of the CineEmbed project is on track: all three pre-registered hypotheses (H1, H2, H3) have been validated, six models have been trained and evaluated against three orthogonal label axes, and the deferred work for the final report has been scoped and prioritized.

- **Data engineering:** 564-dimensional feature matrix over 329,044 films, organized into seven block-contiguous modalities, frozen at v1.2.
- **Modeling:** six runs trained spanning three model tiers (non-deep baselines, simple deep, multi-modal deep + DEC fine-tune); evaluation completed against **primary_genre**, **decade_bin**, and **lang_top10**.
- **Hypotheses:** H1 (DEC > AE on **genre_NMI**), H2 (best-deep $\geq 1.10\times$ best-non-deep), H3 (NMI > 0.15 floor) all PASS; H2 by an order of magnitude.

+205%

relative gain on **genre_NMI** of the best deep model (dec_z64_k21: 0.332) over the strongest non-deep baseline (kmeans_raw_k21: 0.109). Pre-registered threshold for H2 was +10%.

Hypothesis tracker:

ID	Statement	Result	Status
H1	DEC genre_NMI > AE genre_NMI	$0.332 > 0.328$	COMPLETE PASS
H2	Best-deep genre_NMI $\geq 1.10\times$ best-non-deep	+205%	COMPLETE PASS
H3	Best-deep genre_NMI > 0.15 absolute floor	$0.332 \gg 0.15$	COMPLETE PASS

1 Project Overview

1.1 Goal

The CineEmbed project trains unsupervised 64-dimensional representations of movie metadata via multi-modal autoencoder and Deep Embedded Clustering (DEC) architectures, with the objective of demonstrating that the learned latent geometry recovers three orthogonal label axes — primary genre, decade, and original language — substantially better than non-deep KMeans baselines.

1.2 Scope

The dataset is a corpus of 329,044 films assembled from TMDb metadata, awards records (Oscar / BAFTA / Cannes), and Wikipedia-derived director biographies. The feature matrix has 564 dimensions organized into seven block-contiguous modalities:

Modality block	Dim	Contents
numerical	6	log_popularity, log_vote_count, runtime_norm, vote_average_norm, has_vote, has_engagement
genre	22	21-way one-hot + has_genre flag
language	31	top-30 original_language one-hot + flag (sparse: ~99% zero per row)
decade	2	decade_norm + has_release_date
awards	6	log-counts of prior Oscar / BAFTA / Cannes wins and nominations
text	384	all-MiniLM-L6-v2 sentence embeddings of overview + title
director	113	bio_pca_64 + has_director_bio + dir_lang_30 + dir_country_18 + has_director_lang

See Fig. 1 for the long-tailed language distribution that motivates the language block’s sparsity-aware treatment.

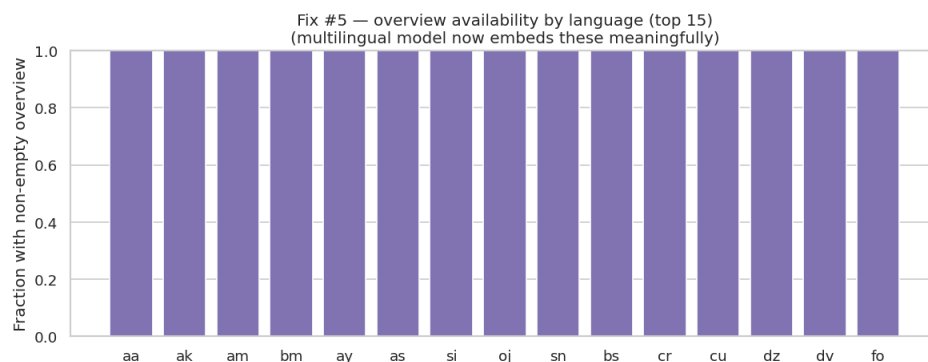


Figure 1: Multilingual coverage of the 329,044-film corpus. Top-3 languages dominate; the long tail is what motivates the modality-specific projection design (Section 3).

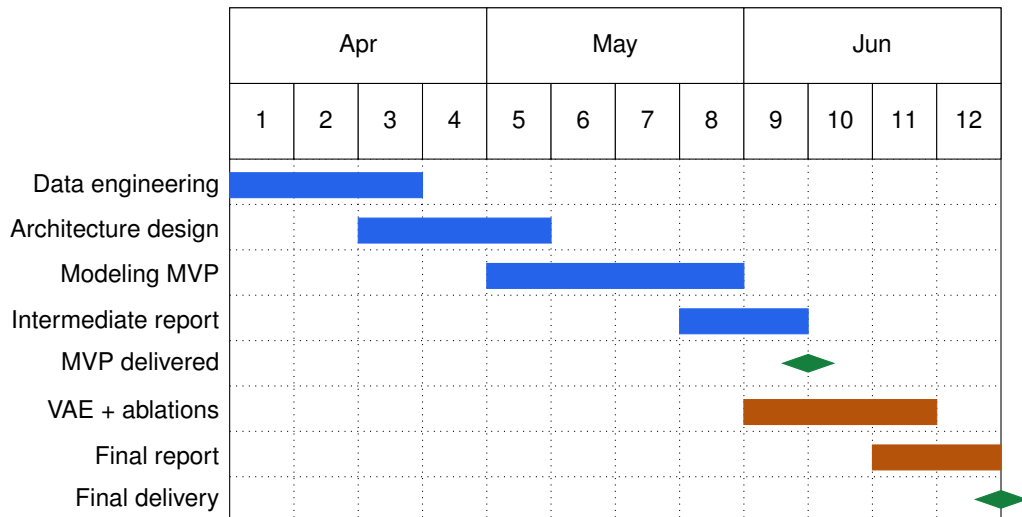
1.3 Team and Responsibilities

Member	Primary responsibility
Ahmet Baran Dincoguz	Architecture design, modeling pipeline, evaluation, repository ownership
Arda Arvas	Data engineering, awards merge, feature blocks
Bertan Kaan Kaya	Director-bio extension, UMAP analysis, presentation deliverables

2 Schedule and Milestones

2.1 Project Timeline

The project is organized into four sequential phases, of which three have completed and one (Final Report) is upcoming. The Gantt chart below shows the planned and actual schedule.



2.2 Milestone Status

Milestone	Target window	Status
Feature matrix v1.2 frozen	Apr 2026	COMPLETE
Multi-modal architecture finalized	Apr 2026	COMPLETE
Six runs trained and evaluated	May 2026	COMPLETE
Pre-registered hypotheses tested	May 2026	COMPLETE
UMAP latent analysis	May 2026	COMPLETE
Intermediate progress report	May 2026	IN PROGRESS
VAE family training ($z = 32/64/128$)	Jun 2026	PLANNED
F1 / F2 modality ablations	Jun 2026	PLANNED
DEC k-sweep (9-cell grid)	Jun 2026	PLANNED
Final report	Jun 2026	PLANNED

3 Work Completed to Date

Three phases have completed since project kickoff. Each subsection below summarizes the phase scope, the deliverables produced, and the empirical evidence supporting completion.

3.1 Data engineering phase

COMPLETE

The team assembled a unified feature matrix from three input sources:

- **TMDB** — 329,044 films, basic metadata (title, overview, runtime, language, genres, votes, popularity, release date).
- **Awards records** — Oscar, BAFTA, Cannes wins and nominations, merged on title and release year, used to compute prior log-counts as the **awards** block.
- **Wikipedia director bios** — raw text scraped per credited director, embedded with `all-MiniLM-L6-v2` [1], then projected to 64 components via PCA. Coverage is partial: 96.8% of films lack a Wikipedia bio for any of their credited directors.

Three engineering decisions are worth noting because they shape the modeling phase:

1. Sparse modalities (language, director-language, director-country) preserved as one-hot blocks rather than embedded, to keep them interpretable.
2. Missing release dates encoded as a binary flag (`has_release_date`) rather than imputed — this turned out to be structurally relevant (see Section 3.3).
3. Director-bio reconstruction loss masked by `has_director_bio` (G2 masking) so that 96.8% of films contribute zero loss for that block, preventing the autoencoder from learning to predict zeros from zeros.

Evidence: The frozen 564-dim feature matrix, EDA notebook (`notebooks/01_eda.ipynb`), and 21 supporting figures under `artifacts/figures/`.

3.2 Architecture design phase

COMPLETE

The team specified and implemented three model heads (deterministic AE, VAE, DEC) on a shared multi-modal backbone. The MVP uses AE and DEC; VAE is implemented but its training is deferred to the final report.

Multi-modal backbone

The encoder applies seven independent `_BlockProjection` layers, one per modality, each compressing its block to a fixed projection dimension before concatenation [2]. The 164-dimensional concatenation feeds a shared backbone — `Linear(164→128) → ReLU → Dropout(0.2) → Linear(128→64)` — yielding the 64-dimensional latent $\mathbf{z}$. Fig. 2 shows the schematic.

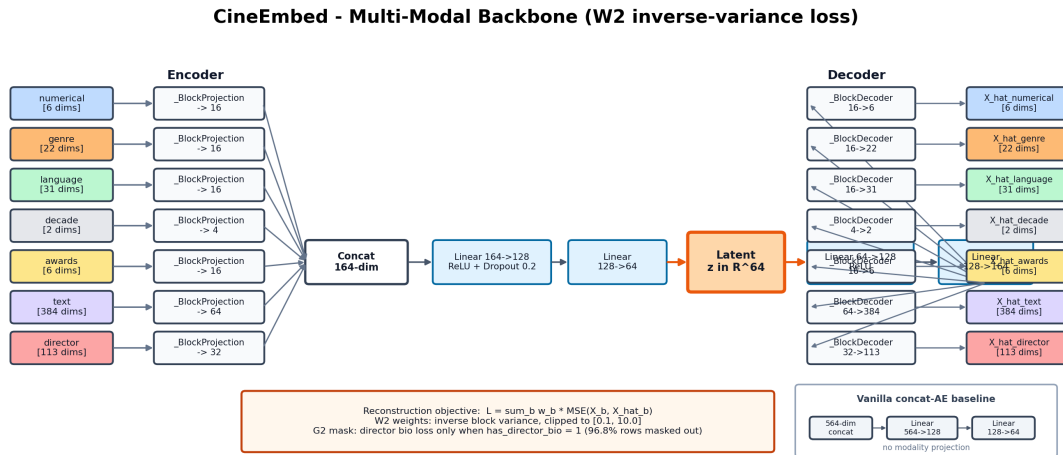


Figure 2: Multi-modal backbone. Seven modality-specific projection blocks reduce each feature group to its `proj_dim`; the resulting 164-dim concatenation feeds a shared backbone with a 64-dim bottleneck. Decoder heads mirror the input structure.

Loss functions

W2 inverse-variance weighting (canonical). Per-block reconstruction losses are reweighted by the inverse of each block’s variance, with clipping for numerical stability:

$$\mathcal{L}_{W2} = \sum_{b=1}^7 w_b \cdot \text{MSE}(X_b, \hat{X}_b), \quad w_b = \text{clip}\left(\frac{1}{\text{var}(X_b)}, 0.1, 10.0\right).$$

The clip bounds $[0.1, 10.0]$ are a peer-review-driven safety measure that prevents the language block (with very low average variance) from dominating early training.

W1 (ablation). Uniform weights $w_b = 1$ for all blocks. Identical architecture and training schedule as W2. This run isolates the contribution of inverse-variance weighting from modality projection.

G2 director-bio masking. Element-wise multiplication of the `bio_pca_64` reconstruction loss by `has_director_bio` so that bio-less rows contribute zero loss for that sub-block.

DEC fine-tuning. Soft assignments $q_{ij} = (1 + \|z_i - \mu_j\|^2)^{-1}$ over $k = 21$ centroids initialized via KMeans++ on pre-trained AE latents; auxiliary target $p_{ij} = q_{ij}^2 / \sum_i q_{ij}$ computed per batch (10× speedup over full-dataset target with no measurable quality regression) [3, 4]. Total loss $\gamma \cdot \text{KL}(P \| Q) + \mathcal{L}_{\text{recon}}$ with $\gamma = 0.1$.

Evidence: Implementation in `src/cineembed/`, design rationale in `docs/adr/0001-modeling-hybrid-architecture.md` (decisions D1–D10).

3.3 Modeling MVP phase

COMPLETE

Six models were trained and evaluated against three orthogonal label axes (`primary_genre`, `decade_bin`, `lang_top10`). All four deep models share a single training recipe: AdamW ($\text{lr} = 10^{-3}$, $\text{weight_decay} = 10^{-5}$), early stopping ($\text{patience} = 10$, $\Delta_{\min} = 10^{-4}$ on `val_loss`), 90/10 random split ($\text{seed}=42$), batch size 1024, on a single Colab T4 GPU. Total wallclock for all six runs was approximately four hours. Implementation in PyTorch [5]; KMeans / NMI / ARI from scikit-learn [6].

Three-axis evaluation methodology

For each trained model, the full-data 64-dim latent is clustered with KMeans ($k=21$, $n_{\text{init}}=20$, $\text{seed}=42$) and the resulting partition is independently scored with NMI (information overlap, label-permutation invariant) and ARI (cluster agreement, harder to satisfy) against the three axes (Fig. 3). No axis is privileged.

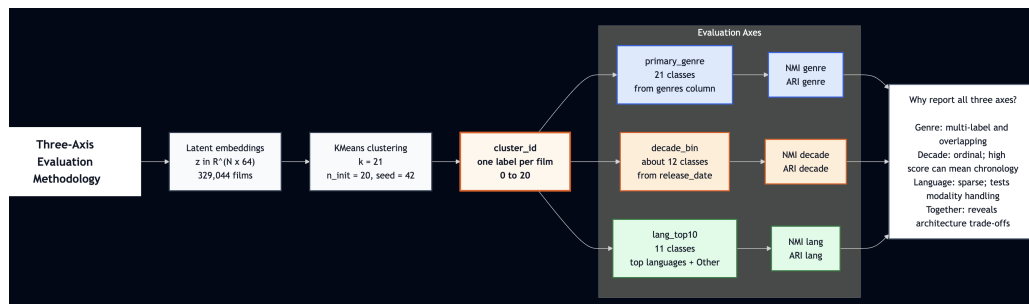


Figure 3: Three-axis evaluation methodology. The single 64-dim latent is clustered once with KMeans ($k=21$); the resulting partition is independently scored with NMI/ARI against three orthogonal label axes.

Main results

Table 1 reports all six runs across all six ($\text{axis} \times \text{metric}$) cells. Six column winners are split across four different models — no architecture dominates — which is the principled-trade-off result documented in Section 5.

Table 1: Main comparison: six runs evaluated against three orthogonal label axes (NMI and ARI per axis). All deep models use $z = 64$; KMeans variants use $k = 21$. Bold cells are column winners.

Tier	Run	gNMI	gARI	dNMI	dARI	INMI	IARI
Non-deep	kmeans_raw_k21	0.109	0.063	0.233	0.093	0.075	0.026
Non-deep	pca_kmeans_k21	0.084	0.061	0.224	0.085	0.094	0.042
Simple deep	vanilla_ae_z64	0.287	0.247	0.369	0.175	0.095	0.030
Ablation (W1)	ae_z64_w1	0.165	0.094	0.367	0.176	0.070	0.026
Multi-modal	ae_z64	0.328	0.229	0.341	0.211	0.264	0.090
Best (DEC)	dec_z64_k21	0.332	0.244	0.342	0.210	0.294	0.090

Headline result — H2

+205%

relative gain on `genre_NMI` of `dec_z64_k21` (0.332) over the strongest non-deep baseline `kmeans_raw_k21` (0.109). Same comparison on `genre_ARI`: +287%. Same comparison on `lang_NMI` vs `pca_kmeans_k21`: +213%.

Architecture-level evidence

Modality projection wins on language (+178%). Multi-modal `ae_z64` produces `lang_NMI` = 0.264 versus the vanilla concat-AE's 0.095, a relative gain of +178%. Fig. 4 shows distinct language micro-clusters in the multi-modal latent.

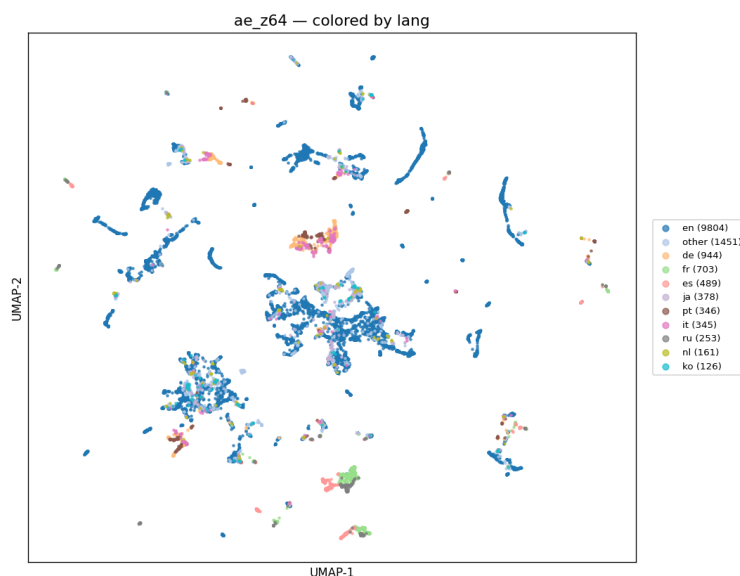


Figure 4: UMAP projection of multi-modal `ae_z64` latents, colored by `lang_top10`. Distinct micro-clusters per language family are visible, supporting the `lang_NMI`=0.264 measurement.

Inverse-variance weighting is critical (+99% on `gNMI`, +277% on `INMI`). The W1 ablation (uniform weights) collapses fine-structure into diffuse blobs (Fig. 5); the same architecture under W2 produces dozens of genre-coherent islands (Fig. 6 center). The W1 run also exhibits early-stopping at epoch 37 versus W2's 69, a model-health signal that the optimizer cannot escape modality imbalance.

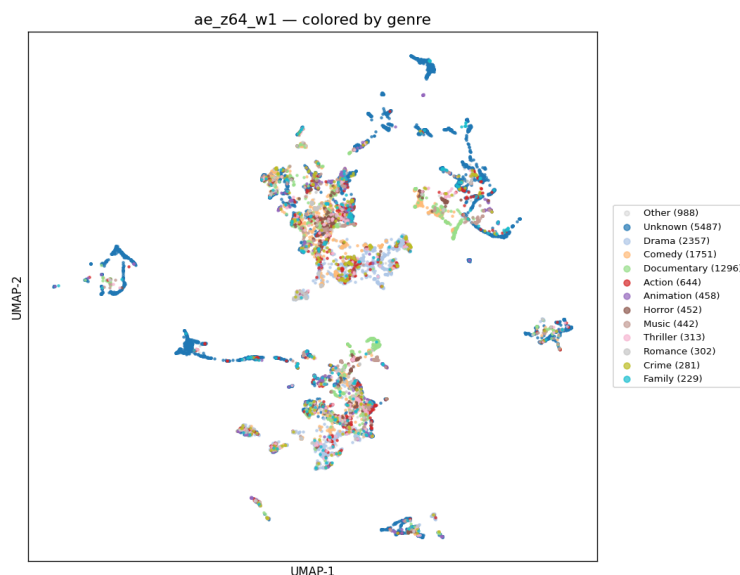


Figure 5: UMAP projection of `ae_z64_w1` (uniform weighting) latents, colored by genre. Diffuse blobs, no fine genre structure — compare with the W2 panel of Fig. 6.

DEC sharpens cluster boundaries (`ARI` > `NMI` gain pattern). DEC was initialized from `ae_z64`'s encoder and trained for 21 epochs of KL+reconstruction. `genre_NMI` improves by +1.2% but `genre_ARI` improves by +6.6% and `lang_NMI` by +11.4%. The diagnostic signature is that `ARI` gain > `NMI` gain:

information content is similar between the AE and DEC latents, but DEC's partition is more crisp. Cluster health was monitored throughout: `total_reinit = 0` across all 21 DEC epochs, meaning all 21 KMeans++ centroids survived KL training without collapse.

Latent-topology evolution

Figure 6 captures the architecture progression visually: vanilla concat-AE produces two mega-blobs, multi-modal `ae_z64` produces dozens of small genre-coherent islands, and `dec_z64_k21` produces even tighter atomized islands with sharper inter-cluster gaps.

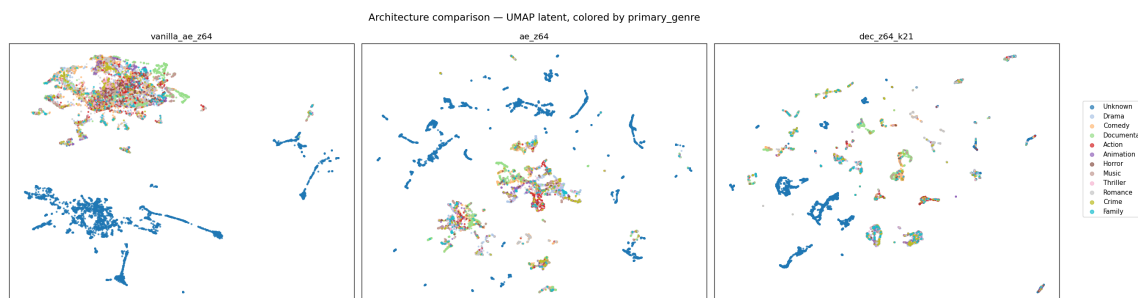


Figure 6: Latent topology comparison (UMAP projection [7], 15K subsample, cosine metric). Left: `vanilla_ae_z64` (two mega-blobs). Center: multi-modal `ae_z64` (dozens of islands). Right: `dec_z64_k21` (tighter atomized islands). All three panels use the same projection settings; only the underlying 64-dim latent differs.

Bonus finding — missing-data sub-manifold

Coloring the DEC latent by `decade_bin` reveals an unexpected pattern (Fig. 7): films with `decade_bin = 0` (the ~7.4% of the corpus with missing release dates) form an isolated cluster in the upper-right of the latent. The same red points are visible in vanilla and multi-modal decade plots, but DEC's KL pressure compresses them into the cleanest partition. Importantly, the model was not forced to encode missingness as latent geometry — `has_release_date` is one binary input among 564. Yet across all four deep architectures, "no release date" emerges as a dimension of latent geometry, not just a flag. This finding was not predicted by the pre-registered hypotheses and is recorded as a representation-learning interpretability win.

Evidence: Six trained checkpoints under `artifacts/checkpoints/`, full numerical results in `artifacts/eval/results.json`, 13 UMAP visualizations under `artifacts/figures/umap/`, and reproducibility notebook `notebooks/06_umap.ipynb`.

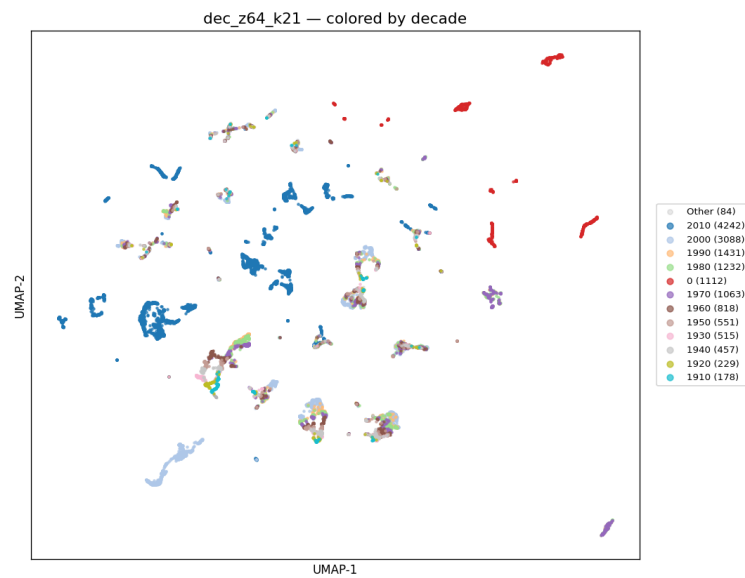


Figure 7: UMAP projection of `dec_z64_k21` latents, colored by `decade_bin`. The isolated red cluster in the upper-right corresponds to films with missing release date — a coherent sub-manifold that emerged without being engineered.

4 Work In Progress

Two workstreams are active during the intermediate-report window:

Workstream	Description	Status
This document	Compiling the intermediate progress report (LaTeX), the source-of-truth tables, and the presentation deck.	IN PROGRESS
UMAP analysis writeup	Documenting the latent-topology and missing-data findings (Section 3.3) for inclusion in the final report's interpretability section.	IN PROGRESS

Both are expected to complete by 2026-05-09.

5 Plan to the Final Report

The intermediate phase deliberately scopes the modeling MVP to the strongest set of empirical claims achievable within the timeline. The following items are deferred to the final report; each has an owner and a target completion window.

Item	Why deferred	Target	Owner
VAE family ($z = 32, 64, 128$)	Probabilistic head implemented but not trained; needed for AE-vs-VAE comparison.	2026-06-05	A.B. Dincoguz
AE latent-dim sweep ($z = 32, 128$)	D2 fixed $z = 64$ for the MVP; sweep informs the final hyperparameter discussion.	2026-06-05	A.B. Dincoguz
F1 modality ablation (no text)	Quantifies the contribution of the 384-dim sentence-embedding block.	2026-06-08	A. Arvas
F2 modality ablation (no director profile)	Quantifies the value of the (sparse, 96.8%-missing) Wikipedia bio block.	2026-06-08	B.K. Kaya
DEC k-sweep (9-cell grid $z \times k$)	MVP runs only the $z=64 \times k=21$ cell.	2026-06-10	A.B. Dincoguz
W4 (Kendall learned uncertainty)	Optional stretch loss; per-block weights as parameters rather than fixed.	2026-06-12	A.B. Dincoguz
Linear probing on frozen latents	Supplementary evaluation against held-out classifiers per axis.	2026-06-12	A. Arvas
5-seed confidence intervals	MVP results are single-seed; CIs needed for the final report's robustness claim.	2026-06-14	B.K. Kaya
Full reproducibility audit	Single deterministic run script that reproduces all numbers from a frozen feature matrix.	2026-06-15	A.B. Dincoguz

The final report submission target is **2026-06-16**, leaving one day of buffer for typesetting and integration.

6 Risks and Mitigation

Risk 1 — Compute budget for the deferred VAE / k-sweep workload

The deferred experiments add roughly 9 additional training runs (3 VAE-z plus 6 DEC-k cells beyond the MVP). At the MVP's observed wallclock of ~40 minutes per deep run, this is 6 GPU-hours — within Colab's free-tier daily quota but tight. **Mitigation:** batch experiments overnight and use checkpointing so any single run can be resumed if Colab disconnects.

Risk 2 — Director-bio coverage limits F2 ablation interpretability

With 96.8% of films missing a Wikipedia director bio, the F2 ablation (no-director-profile) tests removal of a block that is mostly already masked away by G2. The expected effect size is small. **Mitigation:** explicitly report the ~3.2% sub-population effect size separately and document the limitation in the final-report results section.

Risk 3 — Single-seed results may overstate stability

MVP results are computed from a single seed (42). The 5-seed CI work is scheduled but if it produces wide intervals, some headline gains may need softer wording. **Mitigation:** run the 5-seed sweep early (target 2026-06-04, ahead of the 2026-06-14 deadline) so results can be reframed if needed before the final write-up.

7 Deliverables Status

Deliverable	Location	Status
Frozen 564-dim feature matrix v1.2	artifacts/features/	COMP
Six trained model checkpoints	artifacts/checkpoints/	COMP
Evaluation results JSON	artifacts/eval/results.json	COMP
13 UMAP visualizations	artifacts/figures/umap/	COMP
Findings document	docs/FINDINGS.md	COMP
Architecture decision record (D1–D10)	docs/adr/0001-modeling....md	COMP
Intermediate progress report (this doc)	docs/report/intermediate-progress-report.pdf	IN PRO
Intermediate presentation deck	docs/presentation/intermediate-progress-presentation.pptx	IN PRO

A Per-Run Numerical Results

Reproduced verbatim from `artifacts/eval/results.json` at 3-decimal precision.

Run	gNMI	gARI	dNMI	dARI	INMI	lARI	epochs	val_loss
vanilla_ae_z64	0.287	0.247	0.369	0.175	0.095	0.030	58	0.013
ae_z64	0.328	0.229	0.341	0.211	0.264	0.090	69	0.021
ae_z64_w1	0.165	0.094	0.367	0.176	0.070	0.026	37	0.045
dec_z64_k21	0.332	0.244	0.342	0.210	0.294	0.090	21	0.127
kmeans_raw_k21	0.109	0.063	0.233	0.093	0.075	0.026	—	—
pca_kmeans_k21	0.084	0.061	0.224	0.085	0.094	0.042	—	—

The DEC `val_loss` of 0.127 combines reconstruction and KL terms and is not directly comparable to the AE pure-reconstruction `val_loss` values.

B Decision Log Summary

The complete decision rationale is in `docs/adr/0001-modeling-hybrid-architecture.md`. One-line summaries:

ID	Decision
D1	Hybrid C-structure $\times$ D-architecture: modality-specific projections + shared backbone.
D2	Latent dim $z = 64$ for the MVP ($z = 32$ and $z = 128$ deferred to the final report).
D3	W2 inverse-variance weighting + W1 ablation; W4 deferred as optional stretch.
D4	G2 director-bio masking (96.8% of films lack a Wikipedia bio).
D5	Per-model notebooks plus a shared <code>src/cineembed</code> package.
D6	Tier-2 evaluation: 3-axis NMI/ARI as core; linear probing deferred.
D7	L4 ground truth: three-axis (genre, decade, language) labels.
D8	DEC $k = 21$; k-sweep deferred.
D9	Peer-review fixes: 3 baselines, weight clipping, β warmup, relative criteria.
D10	Batch-wise DEC P target rather than full-dataset ($10\times$ speedup, no quality regression).

References

- [1] Nils Reimers and Iryna Gurevych. “Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks”. In: *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 2019.
- [2] Yury Gorishniy et al. “Revisiting Deep Learning Models for Tabular Data”. In: *Advances in Neural Information Processing Systems 34 (NeurIPS)*. 2021.
- [3] Junyuan Xie, Ross Girshick, and Ali Farhadi. “Unsupervised Deep Embedding for Clustering Analysis”. In: *Proceedings of the 33rd International Conference on Machine Learning (ICML)*. 2016, pp. 478–487.
- [4] Xifeng Guo et al. “Improved Deep Embedded Clustering with Local Structure Preservation”. In: *Proceedings of the 26th International Joint Conference on Artificial Intelligence (IJCAI)*. 2017, pp. 1753–1759.
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- [6] F. Pedregosa et al. “Scikit-learn: Machine Learning in Python”. In: *Journal of Machine Learning Research* 12 (2011), pp. 2825–2830.
- [7] Leland McInnes, John Healy, and James Melville. “UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction”. In: *arXiv preprint arXiv:1802.03426* (2018).